Artificial Intelligence & Machine Learning

Sample Knowledge Document for Mini-RAG

1. Introduction to Artificial Intelligence

Artificial Intelligence (AI) is the field of computer science focused on building systems that can perform tasks that normally require human intelligence. These tasks include understanding language, recognizing images, making predictions, planning actions, and solving problems. Modern AI systems can use data and algorithms to identify patterns and produce useful outputs.

2. Machine Learning

Machine Learning (ML) is a major area of AI in which computers learn patterns from data instead of relying entirely on explicitly written rules. A machine learning model is trained using examples and then used to make predictions or decisions on new data.

For example, a spam classifier can learn from previously labeled emails. During training, the model learns characteristics associated with spam and legitimate messages. When a new email arrives, the trained model estimates which category it belongs to.

3. Types of Machine Learning

Supervised learning: The model learns from labeled examples. Common tasks include classification and regression. Classification predicts categories, while regression predicts numerical values.

Unsupervised learning: The model works with data that does not have target labels. Clustering is a common example, where similar observations are grouped together.

Reinforcement learning: An agent learns by interacting with an environment and receiving rewards or penalties. The goal is to learn actions that maximize long-term reward.

4. Deep Learning

Deep Learning is a branch of machine learning based on neural networks with multiple layers. Deep learning models can automatically learn complex representations from large amounts of data. Convolutional Neural Networks (CNNs) are widely used for image tasks, while Transformer architectures are important for language and multimodal applications.

5. Natural Language Processing

Natural Language Processing (NLP) focuses on enabling computers to work with human language. NLP applications include text classification, sentiment analysis, translation, question answering, summarization, and information extraction.

6. Generative AI

Generative AI refers to models that can create new content based on patterns learned from training data. Large Language Models (LLMs) are generative AI systems designed to process and generate natural language. They can be used for question answering, summarization, coding assistance, and conversational applications.

7. Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation, commonly called RAG, combines information retrieval with a generative language model. Instead of relying only on information stored in the model, a RAG system first retrieves relevant information from an external knowledge source and then provides that information to the language model as context.

A typical RAG pipeline contains several stages: document ingestion, text extraction, chunking, embedding generation, vector storage, similarity search, context selection, and answer generation. This approach is useful for question answering over private or specialized documents.

8. Embeddings and Vector Search

An embedding is a numerical representation of text or another type of data. Similar pieces of information tend to have embeddings that are close together in vector space. Vector databases and libraries such as FAISS can perform similarity searches to retrieve the chunks most relevant to a user's question.

9. AI Model Evaluation

Machine learning models should be evaluated using metrics appropriate to their task. Classification can use accuracy, precision, recall, and F1-score. Regression can use metrics such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). For RAG systems, evaluation can also consider retrieval relevance, factuality, and answer quality.

10. Key Takeaways

AI is the broader field of creating intelligent computer systems. Machine Learning allows systems to learn patterns from data. Deep Learning uses multi-layer neural networks for complex tasks. Generative

AI creates new content, while RAG connects generative models with external knowledge sources. Together, these technologies can be used to build powerful document question-answering applications.